مكمل الفيرست

biology 102

chapter 23

The Evolution of Populations

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Consept 23.1

1) Which of the following is the best modern definition of evolution?
A) descent with modification
B) change in the number of genes in a population over time
C) survival of the fittest
D) inheritance of acquired characters
Answer: A

2) Microevolutions occur when ________.
A) a bird has a beak of a particular size that does not grow larger during a drought

B) changes in allele frequencies in a population occur over generations
C) gene flow evenly transfers alleles between populations
D) individuals within all species vary in their phenotypic traits
Answer: B

3) Which statement about the beak size of finches on the island of Daphne Major during
prolonged drought is true?
A) Each bird evolved a deeper, stronger beak as the drought persisted.
B) Each bird's survival was strongly influenced by the depth and strength of its beak as the
drought persisted.
C) Each bird that survived the drought produced only offspring with deeper, stronger beaks than
seen in the previous generation.
D) The frequency of the strong-beak alleles increased in each bird as the drought persisted.
Answer: B

4) Which statement about variation is true?
A) All phenotypic variation is the result of genotypic variation.
B) All genetic variation produces phenotypic variation.
C) All nucleotide variability results in neutral variation.
D) All new alleles are the result of nucleotide variability.
Answer: D

5) Which of the following descriptions illustrates phenotype variation caused by environment?
A) inheritance of body builder "physique"
B) diet of caterpillars changes their morphology
C) variation in horse coat color
D) average beak depth during drought

Answer: B

6) Genetic variation ________.
A) is created by the direct action of natural selection
B) arises in response to changes in the environment
C) must be present in a population before natural selection can act upon the population
D) tends to be reduced when diploid organisms produce gametes
Answer: C

7) HIV's genome of RNA includes the code for reverse transcriptase (RT), an enzyme that acts
early in infection to synthesize a DNA genome off of an RNA template. The HIV genome also
codes for protease (PR), an enzyme that acts later in infection by cutting long viral polyproteins
into smaller, functional proteins. Both RT and PR represent potential targets for antiretroviral
drugs. Drugs called nucleoside analogs (NA) act against RT, whereas drugs called protease
inhibitors (PI) act against PR.
Which of the following treatment options would most likely avoid the evolution of drug-resistant
HIV (assuming no drug interactions or side effects)?
A) Use a series of NAs, one at a time, and change about once a week.
B) Use a single PI, but slowly increase the dosage over the course of a week.
C) Use high doses of NA and a PI at the same time for a period not to exceed one day.
D) Use moderate doses of NA and two different PIs at the same time for several months.
Answer: D

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drugs. Drugs called nucleoside analogs (NA) act against RT, whereas drugs called protease
inhibitors (PI) act against PR.
Which mechanism produces variation for evolution by shuffling existing alleles?
A) rapid reproduction
B) sexual reproduction
C) mutation
D) changes in chromosome numbers
Answer: B

9) The following experiment is used for the corresponding question.
A researcher discovered a species of moth that lays its eggs on oak trees. Eggs are laid at two
distinct times of the year: early in spring when the oak trees are flowering and in midsummer
when flowering is past. Caterpillars from eggs that hatch in spring feed on oak flowers and look
like oak flowers. But caterpillars that hatch in summer feed on oak leaves and look like oak
twigs.
How does the same population of moths produce such different-looking caterpillars on the same
trees? To answer this question, the biologist caught many female moths from the same

population and collected their eggs. He put at least one egg from each female into eight identical cups. The eggs hatched, and at least two larvae from each female were maintained in one of the four temperature and light conditions listed below.

Temperature	Day Length
Springlike	Springlike
Springlike	Summerlike
Summerlike	Springlike
Summerlike	Summerlike

In each of the four environments, one of the caterpillars was fed oak flowers, the other oak leaves. Thus, there were a total of eight treatment groups (4 environments × 2 diets).
Which of the following is a testable hypothesis that would explain the differences in caterpillar appearance observed in this population?
A) The longer day lengths of summer trigger the development of twig-like caterpillars.
B) Winter causes ugly caterpillar and trees.
C) Differences in air pressure, due to differences in elevation, trigger the development of different types of caterpillars.
D) Differences in diet trigger the development of different types of caterpillars.
Answer: D

Concept 23.2

10) Cystic fibrosis is a genetic disorder in homozygous recessives that causes death during the

teenage years. If 9 in 10,000 newborn babies have the disease, what are the expected frequencies
of the dominant (*A1*) and recessive (*A2*) alleles according to the Hardy-Weinberg equation?
A) f(*A1*) = 0.9997, f(*A2*) = 0.0003
B) f(*A1*) = 0.9800, f(*A2*) = 0.0200
C) f(*A1*) = 0.9700, f(*A2*) = 0.0300
D) f(*A1*) = 0.9604, f(*A2*) = 0.0392
Answer: C
11) Suppose 64% of a remote mountain village can taste phenylthiocarbamide (PTC) and must,
therefore, have at least one copy of the dominant PTC taster allele. If this population conforms to
Hardy-Weinberg equilibrium for this gene, what percentage of the population must be
heterozygous for this trait?
A) 16%
B) 32%
C) 40%
D) 48%
Answer: D
12) If individuals tend to mate within a subset of the population, there is ________.
A) no selection
B) no genetic drift
C) no gene flow
D) random mating
Answer: D

13) Which Hardy-Weinberg condition is affected by population size?
A) selection
B) genetic drift
C) gene flow

D) no mutation
Answer: B

14) Use the following information to answer the question below.
Researchers studying a small milkweed population note that some plants produce a toxin and
other plants do not. They identify the gene responsible for toxin production. The dominant allele
(*T*) codes for an enzyme that makes the toxin, and the recessive allele (*t*) codes for a
nonfunctional enzyme that cannot produce the toxin. Heterozygotes produce an intermediate
amount of toxin. The genotypes of all individuals in the population are determined (see chart)
and used to determine the actual allele frequencies in the population.

Genotype Frequencies			Allele Frequencies	
TT	*Tt*	*tt*	*T*	*T*
0.56	0.28	0.16		

Is this population in Hardy-Weinberg equilibrium?
A) Yes.
B) No; there are more heterozygotes than expected.
C) No; there are more homozygotes than expected.
D) More information is needed to answer this question.
Answer: C

15) Which one of the following conditions would allow gene frequencies to change by chance?
A) large population
B) small populations
C) mutation
D) gene flow
Answer: B

16) The higher the proportion of loci that are "fixed" in a population, the lower are that
population's ________.
A) nucleotide variability
B) chromosome number
C) average heterozygosity
D) nucleotide variability and average heterozygosity
Answer: D

17) Whenever diploid populations are in Hardy-Weinberg equilibrium at a particular locus,
________.
A) the allele's frequency should not change from one generation to the next
B) natural selection, gene flow, and genetic drift are acting equally to change an allele's
frequency
C) two alleles are present in equal proportions
D) individuals within the population are evolving
Answer: A

18) In the formula for determining a population's genotype frequencies, the "*2*" in the term *2pq* is
necessary because ________.
A) the population is diploid
B) heterozygotes can come about in two ways
C) the population is doubling in number
D) heterozygotes have two alleles
Answer: B

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Answer: D

20) In a Hardy-Weinberg population with two alleles, *A* and *a*, that are in equilibrium, the
frequency of the allele *a* is 0.3. What is the frequency of individuals that are homozygous for this
allele?
A) 0.09
B) 0.49
C) 0.9
D) 9.0
Answer: A

21) In a Hardy-Weinberg population with two alleles, *A* and *a*, that are in equilibrium, the
frequency of allele *a* is 0.2. What is the frequency of individuals that are heterozygous for this
allele?
A) 0.020
B) 0.04
C) 0.16
D) 0.32
Answer: D

22) In a Hardy-Weinberg population with two alleles, *A* and *a*, that are in equilibrium, the
frequency of allele *a* is 0.1. What is the frequency of individuals with *AA* genotype?
A) 0.20
B) 0.32
C) 0.42
D) 0.81
Answer: D

23) You sample a population of butterflies and find that 56% are heterozygous at a particular
locus. What should be the frequency of the homozygous individuals in this population?
A) 0.08
B) 0.09
C) 0.70
D) 0.50
Answer: D

24) In peas, a gene controls flower color such that *R* = purple and *r* = white. In an isolated pea
patch, there are 36 purple-flowering plants and 64 white-flowering plants. Assuming Hardy-
Weinberg equilibrium, what is the value of *q* for this population?
A) 0.36
B) 0.64
C) 0.75
D) 0.80
Answer: D

25) A large population of laboratory animals has been allowed to breed randomly for a number
of generations. After several generations, 25% of the animals display a recessive trait (*aa*), the
same percentage as at the beginning of the breeding program. The rest of the animals show the
dominant phenotype, with heterozygotes indistinguishable from the homozygous dominants.
What is the most reasonable conclusion that can be drawn from the fact that the frequency of the
recessive trait (*aa*) has not changed over time?

A) The two phenotypes are about equally adaptive under laboratory conditions.
B) The genotype *AA* is lethal.
C) There has been a high rate of mutation of allele *A* to allele *a*.
D) There has been sexual selection favoring allele *a*.
Answer: A

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of generations. After several generations, 25% of the animals display a recessive trait (*aa*), the
same percentage as at the beginning of the breeding program. The rest of the animals show the
dominant phenotype, with heterozygotes indistinguishable from the homozygous dominants.
What is the estimated frequency of allele *A* in the gene pool?
A) 0.25
B) 0.50
C) 0.75
D) 0.125
Answer: B

27) A large population of laboratory animals has been allowed to breed randomly for a number
of generations. After several generations, 25% of the animals display a recessive trait (*aa*), the
same percentage as at the beginning of the breeding program. The rest of the animals show the
dominant phenotype, with heterozygotes indistinguishable from the homozygous dominants.
What proportion of the population is probably heterozygous (*Aa*) for this trait?
A) 0.05

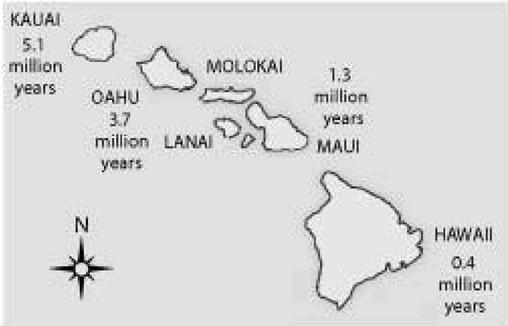
B) 0.25
C) 0.50
D) 0.75
Answer: C

Concept 23.3

28) Which one of these processes describes bottleneck effect?
A) chance events that change allele frequency
B) alleles transferred to the next generation in portions that differ from previous generation
C) transfer of alleles in and out of a population due to movement of fertile individuals
D) sudden change in environments that alters gene frequency of a population
Answer: D

29) Comparisons of Neanderthal DNA revealed that there are more similarities to non-African
DNA than reference sequences from West Africans. Additionally, scientists found that
Neanderthal DNA is as closely related to East Asians as to Europeans. This indicates that
interbreeding occurred before human migration further east. What process of population genetics
generated these results?
A) adaptive evolution
B) gene flow
C) gene drift
D) nonrandom mating
Answer: B

30)



Soon after the island of Hawaii rose above the sea surface (somewhat less than one million years ago), the evolution of life on this new island should have been most strongly influenced by ________.
A) a genetic bottleneck
B) sexual selection
C) habitat differentiation
D) the founder effect
Answer: D

31) In 1983, a population of dark-eyed junco birds became established on the campus of the University of California, San Diego (UCSD), which is located many miles from the junco's normal habitat in the mixed-coniferous temperate forests in the mountains. Juncos have white outer tail feathers that the males display during aggressive interactions and during courtship displays. Males with more white in their tail are more likely to win aggressive interactions, and females prefer to mate with males with more white in their tails. Females have less white in their tails than do males, and display it less often. (Pamela J. Yeh. 2004. Rapid evolution of a sexually selected trait following population establishment in a novel habitat. *Evolution* 58[1]:166-74.)

The UCSD campus male junco population tails were, on average, 36% white, whereas the tails of
males from nearby mountain populations averaged 40-45% white. If this observed trait
difference were due to a difference in the original colonizing population, it would most likely be
due to ________.
A) mutations in the UCSD population
B) gene flow between populations
C) a genetic bottleneck
D) a founder effect
Answer: D

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selected trait following population establishment in a novel habitat. *Evolution* 58[1]:166-74.)
The UCSD campus male junco population tails are about 36% white, whereas the tails of males
from nearby mountain populations are about 40-45% white. The founding stock of UCSD birds

was likely from the nearby mountain populations because some of those birds overwinter on the
UCSD campus each year. Population sizes on the UCSD campus have been reasonably large,
and there are significant habitat differences between the UCSD campus and the mountain
coniferous forests; UCSD campus has a more open environment (making birds more visible) and
a lower junco density (decreasing intraspecific competition) than the mountain forests. Given
this information, which of the following evolutionary mechanisms do you think is most likely
responsible for the difference between the UCSD and mountain populations?
A) natural selection
B) genetic drift
C) gene flow
D) mutation
Answer: A

33) The Dunkers are a religious group that moved from Germany to Pennsylvania in the mid-
1700s. They do not marry with members outside their own immediate community. Today, the
Dunkers are genetically unique and differ in gene frequencies, at many loci, from all other
populations including those in their original homeland. Which of the following mechanisms
likely explains the genetic uniqueness of this population?
A) population bottleneck and Hardy-Weinberg equilibrium
B) heterozygote advantage and stabilizing selection
C) mutation and natural selection
D) founder effect and genetic drift
Answer: D

34) An earthquake decimates a ground-squirrel population, killing 98% of the squirrels. The
surviving population happens to have broader stripes, on average, than the initial population. If
broadness of stripes is genetically determined, what effect has the ground-squirrel population
experienced during the earthquake?
A) directional selection
B) disruptive selection
C) a founder event
D) a genetic bottleneck
Answer: D

35) Which of the following is the most predictable outcome of increased gene flow between two
populations?
A) lower average fitness in both populations
B) higher average fitness in both populations
C) increased genetic difference between the two populations
D) decreased genetic difference between the two populations
Answer: D

36) In 1986, a nuclear power accident in Chernobyl, USSR (now Ukraine), led to high radiation
levels for miles surrounding the plant. The high levels of radiation caused elevated mutation
rates in the surviving organisms, and evolutionary biologists have been studying rodent
populations in the Chernobyl area ever since. Based on your understanding of evolutionary
mechanisms, which of the following most likely occurred in the rodent populations following the
accident?

A) Mutations caused major changes in rodent physiology over time.
B) Mutation led to increased genetic variation.
C) Mutation caused genetic drift and decreased fitness.
D) Mutation caused the fixation of new alleles.
Answer: B

37) Over time, the movement of people on Earth has steadily increased. This has altered the
course of human evolution by increasing ________.
A) nonrandom mating
B) geographic isolation
C) genetic drift
D) gene flow
Answer: D

38) You are maintaining a small population of fruit flies in the laboratory by transferring the flies
to a new culture bottle after each generation. After several generations, you notice that the
viability of the flies has decreased greatly. Recognizing that small population size is likely to be
linked to decreased viability, the best way to reverse this trend is to ________.
A) cross your flies with flies from another lab
B) reduce the number of flies that you transfer at each generation
C) transfer only the largest flies
D) change the temperature at which you rear the flies
Answer: A